

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Owned ML development from problem formulation through offline evaluation and **A/B testing** to **production rollout**, defining quality and regression metrics that automatically blocked underperforming model releases.
- Implemented **GraphRAG** within a production **retrieval and ranking** pipeline, combining **LLM-derived** semantic signals with graph context to improve relevance on complex, multi-hop queries.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Owned an end-to-end **ranking and retrieval** system, redesigning **candidate generation** with **hybrid dense-sparse retrieval** and **re-ranking**; improved **Recall@3** from **61% to 89%** across a 2K-query evaluation set.
- Designed a **Neo4j** entity-relation graph integrated into the retrieval pipeline, enabling **graph-based candidate expansion** and multi-hop relevance signals across a production system serving **50K+ daily requests**.

University of Texas at Arlington | *Graduate Teaching Assistant*

Aug 2024 – May 2025

- Benchmarked 5+ **multi-agent reinforcement learning** architectures on a multi-GPU **PyTorch** cluster, evaluating **reward design** and robustness under distributed interference.
- Developed multi-modal deep learning models in **PyTorch** with **attention mechanisms** and domain-specific architectures, building **evaluation frameworks** and reproducible research workflows.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production Java** microservices serving **1M+ users** at **99.9% uptime**, owning system design, deployment, monitoring, and failure recovery across distributed services.
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations **60%** in one quarter via consistent retry, timeout, and propagation semantics.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- Built a multi-stage retrieval system combining **Pinecone** vector-based **candidate generation**, **hybrid retrieval**, and relevance-based **re-ranking**, with **LangGraph**-orchestrated query decomposition for complex information needs.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built an ML **evaluation platform** with **rubric-based scoring**, regression detection, hallucination analysis, and drift tracking across model versions, converting subjective quality criteria into measurable release metrics.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** distributed storage engine handling concurrent writes with **linearizable reads** and **log compaction**; companion VPC control plane simulator modeling subnet routing.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, TypeScript, SQL

Ranking & Retrieval: Candidate Generation, Hybrid Dense-Sparse Retrieval, Re-Ranking, Embedding Retrieval, Vector Search (Pinecone), Recall@3

ML & Experimentation: PyTorch, Representation Learning, Attention Mechanisms, Offline Evaluation, A/B Testing, Reinforcement Learning

Production ML: End-to-End Model Ownership, Evaluation Frameworks, Observability, Real-Time Low-Latency Serving

AI Retrieval Infra: GraphRAG, Neo4j Knowledge Graphs

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer